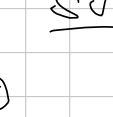


$aba^{-1}b^{-1} = e$
 $ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$

 $S^1 \vee S^2$


 $\frac{S^1 \vee S^1}{\mathbb{Z} \times \mathbb{Z}}$
 $T = \mathbb{Z} \times \mathbb{Z}$
 $T \hookrightarrow \mathbb{Z} \times \mathbb{Z}$
 $\mathbb{Z} \hookrightarrow \mathbb{Z} \times \mathbb{Z}$
 S^1
 $S^1 \vee S^1$

[illegible][illegible]

$\mathbb{R}/\mathbb{P}^2$: 0-cell: 1 pt
 1-cell: 1 pt
 2-cell: 1 pt

$\langle a | a^2 \rangle$

$ab^{-1}b^2 = 0$
 $ab = b^{-1}$
 $a^2 = 1$

Handwritten notes on a grid background:

- Top left: RP
- Top center: $a^2 = 1$
- Top right: $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$
- Middle left: A diagram of a circle with a vertical line segment inside, labeled a at both ends.
- Middle center: RP followed by a list:
 - 1 cell
 - 2 cell
 - 3 cell
- Middle right: A diagram of a circle with a vertical line segment inside, labeled a at both ends.
- Bottom left: A diagram of a circle with a vertical line segment inside, labeled a at both ends.
- Bottom center: $RP = a(RP) = 2.7$
- Bottom right: A diagram of a circle with a vertical line segment inside, labeled a at both ends.

Hand-drawn sketches on graph paper:

- A sphere with a horizontal line passing through its center, and two points on the line.
- A flower-like shape with five petals.
- A complex loop with two dots on it.
- A spiral shape.
- A shape resembling a stylized 'S' or a wave.
- A shape resembling a stylized 'Z' or a zigzag.
- A shape resembling a stylized 'U' or a U-shape.
- A shape resembling a stylized 'V' or a V-shape.
- A shape resembling a stylized 'W' or a W-shape.
- A shape resembling a stylized 'X' or an X-shape.
- A shape resembling a stylized 'Y' or a Y-shape.
- A shape resembling a stylized 'Z' or a Z-shape.
- A shape resembling a stylized 'A' or an A-shape.
- A shape resembling a stylized 'B' or a B-shape.
- A shape resembling a stylized 'C' or a C-shape.
- A shape resembling a stylized 'D' or a D-shape.
- A shape resembling a stylized 'E' or an E-shape.
- A shape resembling a stylized 'F' or an F-shape.
- A shape resembling a stylized 'G' or a G-shape.
- A shape resembling a stylized 'H' or an H-shape.
- A shape resembling a stylized 'I' or an I-shape.
- A shape resembling a stylized 'J' or a J-shape.
- A shape resembling a stylized 'K' or a K-shape.
- A shape resembling a stylized 'L' or an L-shape.
- A shape resembling a stylized 'M' or an M-shape.
- A shape resembling a stylized 'N' or an N-shape.
- A shape resembling a stylized 'O' or an O-shape.
- A shape resembling a stylized 'P' or a P-shape.
- A shape resembling a stylized 'Q' or a Q-shape.
- A shape resembling a stylized 'R' or an R-shape.
- A shape resembling a stylized 'S' or an S-shape.
- A shape resembling a stylized 'T' or a T-shape.
- A shape resembling a stylized 'U' or a U-shape.
- A shape resembling a stylized 'V' or a V-shape.
- A shape resembling a stylized 'W' or a W-shape.
- A shape resembling a stylized 'X' or an X-shape.
- A shape resembling a stylized 'Y' or a Y-shape.
- A shape resembling a stylized 'Z' or a Z-shape.

Hand-drawn sketches on a grid background:

- Top left: A circle with a horizontal line passing through its center, and a small dot above it.
- Top center: A small circle with a horizontal line passing through its center.
- Top right: A large circle with a horizontal line passing through its center, and a small circle with a cross inside it.
- Bottom left: A series of five connected loops, resembling a chain or a spiral.
- Bottom center: A circle with a horizontal line passing through its center, and a small circle with a cross inside it.
- Bottom right: A circle with a horizontal line passing through its center, and a small circle with a cross inside it.
- Far right: A vertical line with a small circle at the top, and a small circle with a cross inside it.

$z = x + iy$
 $\downarrow \text{exp}$
 $K_1 = \frac{1}{4}$
 $K_2 = \frac{1}{4}$
 $2\pi \cdot \frac{1}{4}$

$\int \frac{dx}{x^2} = -\frac{1}{x} + C$